NyayaMitra — an AI lawyer for everyday Bharat

Solution doc for Problem 2. Speak your problem in your own language, get a legal notice you can actually send.

Submitted by	Advait Pathak · AI Generalist
Problem	Problem 2 — AI Lawyer
Prototype	Streamlit front end, FastAPI back end (runs locally or on cloud)
Code	github.com/Advait-Pathak9222/KUKU-FM_IIT-BHU-_Advait_Pathak

⚖️ **TL;DR** — Someone whose landlord kept their deposit, or whose startup stopped paying salary, sends a voice note in Hindi, Marathi or Hinglish. NyayaMitra asks a few follow-ups, calculates exactly what they're owed, checks which laws apply, and drafts a formal legal notice.

It costs about **55 paise per case**, and it passed all **11 end-to-end test scenarios** I wrote, including refusing a domestic-violence case and pointing the user to the right helpline.

Why I picked this problem

Most people in India never go to a lawyer over a small civil dispute: a deposit that wasn't returned, unpaid salary, a faulty product, a freelance cheque that bounced. A consultation costs ₹2,500–5,000, the vocabulary is intimidating, and writing a proper pre-litigation notice is out of reach for most.

Problem 1 deals with English screenplays. Problem 2 is about the same people who listen to Kuku FM: regional-language speakers who would explain their problem in a long, emotional voice note rather than fill in a form. That felt like the more useful thing to build, and the harder one to get right, because the facts, the maths and the law all have to be correct.

Scope. Four kinds of civil recovery: rental deposit, unpaid salary, consumer disputes, and money recovery (including cheque bounce). Criminal matters, domestic violence, dowry, divorce and custody are deliberately out of scope. If any of these come up on any turn, the assistant stops and gives NALSA (15100), the Women's Helpline (1091) or 112.

How it works — System Architecture & Data Flow

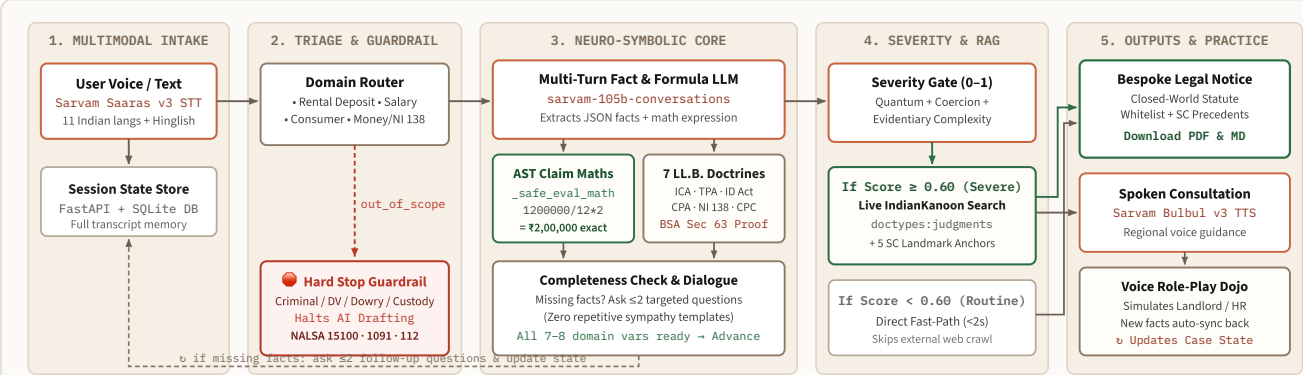


Figure 1: NyayaMitra 5-Stage Neuro-Symbolic Architecture — separating LLM language understanding from deterministic AST arithmetic & statutory guardrails.

1. **Intake and triage.** Sarvam Saaras (`saaras:v3`) transcribes the voice note (11 Indian languages). The conversation is routed to one of the four civil domains, or stopped immediately if it's out of scope.
2. **Claim maths.** People describe money loosely: “18 lakh CTC, 12 fixed, 6 is ESOPs that haven't vested, two months not paid.” The LLM only writes the formula, e.g. `1200000 / 12 * 2` , and a restricted Python AST evaluator (`_safe_eval_math`) computes the exact figure.
3. **Legal check.** The case is tested against seven core LL.B. subjects: Contract Act (Sec 10, 27, 70, 73, 74), TPA Sec 108(m) for wear and tear, ID Act Sec 2(s) and 33C(2) to tell a “workman” from a manager, Consumer Protection Act Sec 2(7) and 34(2)(d) for filing from your home city on e-Daakhil, NI Act Sec 138/139, CPC Order 37 with the Limitation Act, and Sec 63 of the Bharatiya Sakshya Adhiniyam, 2023 for WhatsApp and UPI evidence.
4. **Severity gate.** A 0–1 score based on the amount, pressure tactics by the other side (withheld relieving letter, employment bond, unregistered lease, bounced cheque) and legal complexity. Below 0.60 it answers straight away. At 0.60 or above it searches IndianKanoon.org judgments live and pairs them with known Supreme Court precedents.
5. **Notice and practice.** Facts, maths and precedents go into a formal Legal Notice of Demand (PDF and Markdown). There's also a voice role-play (Sarvam Bulbul `bulbul:v3`) where the user can practise arguing with the landlord or employer, and any new facts they mention get added to the case.

Stopping the model from making things up

A general chatbot in this role will happily invent a section number, get a three-step subtraction wrong, or reach for IPC 420 in what is plainly a civil recovery. So I split the work: the model handles language and judgement, and anything that has to be exactly right is done in code.

What the LLM does

- Keeps track of facts across a multi-turn conversation
- Writes the claim formula (never the final answer)
- Picks up case-specific details and runs the legal diagnosis
- Replies in the user's language without repeating itself
- Plays the other party in adversarial voice role-play

Model: `sarvam-105b-conversations`. A single `call_llm` wrapper means it can be swapped for OpenAI, Gemini or Claude.

What's fixed in code

- **Every rupee figure** comes from the AST evaluator.
- **Citations** are limited to a per-domain whitelist of statutes (ICA, TPA, Payment of Wages Act, ID Act, CPA 2019, NI Act, CPC, BSA 2023) and five vetted precedents: *Fateh Chand*, *Central Inland Water Transport*, *Kailash Nath*, *Rangappa*, *Pioneer Urban*.
- **Escalation** on criminal, DV, dowry, divorce or custody signals is a hard stop in Python, not a prompt instruction.

What one case costs

Step	Model / service	Usage	Cost
Voice transcription	Sarvam Saaras `saaras:v3`	~20 s of audio	₹0.15
Reasoning, maths, drafting	Sarvam-105B-Conversations	~1,800 tokens across the intake	₹0.25
Precedent search	IndianKanoon (HTTP) + Python	Only when severity ≥ 0.60	₹0.00
Spoken advice	Sarvam Bulbul `bulbul:v3`	~15 s of audio	₹0.15
Per case — notice PDF plus spoken consultation			**~₹0.55**

Testing

I ran 11 end-to-end scenarios (all unedited traces in `raw_outputs/`), including a four-turn Hindi salary conversation, wear-and-tear on an unregistered lease, a freelance cheque bounce and live precedent lookup.

Case	Scenario	Domain (severity)	Result	What it showed
`TC 01–04`	Deposit and oral-lease cases in Bengaluru, Pune, Delhi	`rental_deposit`	**Pass**	Sec 108(m) TPA and Sec 63 BSA applied; ₹1,50,000 − ₹65,000 = ₹85,000
`TC 05–07, 09`	4-turn Hindi chat: ₹18L CTC, ₹12L fixed, service bond threat	`unpaid_salary (0.85)`	**Pass**	₹12L ÷ 12 × 2 = ₹2,00,000, ESOPs excluded; cited *Central Inland Water Transport*
`TC 11`	Freelance/business loan + dishonoured HDFC cheque	`money_recovery (0.85)`	**Pass**	15-day statutory demand notice under NI Act 138/139; cited *Rangappa v. Sri Mohan*
`TC 10`	Faulty EV scooter battery with warranty refusal	`consumer_dispute`	**Pass**	CPA Sec 2(10) defect & home-city filing under Sec 34(2)(d); cited *Pioneer Urban*
`TC 08`	Domestic violence, dowry harassment, child custody	`out_of_scope`	**Pass**	Stopped drafting immediately; gave NALSA 15100 and Women's Helpline 1091

What went wrong along the way

I used coding assistants for scaffolding; Sarvam models do the speech and reasoning at runtime. Early builds forgot numbers from earlier turns (the CTC from turn 3 was gone by turn 4), got simple subtraction wrong (₹135k − ₹35k came out as ₹95k), and kept repeating the same sympathy line. Passing the full transcript each turn, moving all maths into the AST evaluator, and adding anti-repetition instructions fixed all three.

Known limitations

- Vague dates like “Diwali ke aas-paas” are stored as said unless the user clarifies.
- Heavy traffic noise can garble unfamiliar English company names in transcription.
- Live IndianKanoon scraping relies on public HTML structure; if offline, the engine falls back to its vetted Supreme Court anchor precedents.